$f(n) \sim g(n)$ iff $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 1$

(Stirling)

$$n! \sim \sqrt{2\pi}\left(\frac{n}{e}\right)^n \sqrt{n} = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

$X = \{r_1, r_2, \dots, r_{26}, b_1, b_2, \dots, b_{26}\}$ $\quad \Omega = \{Y \subseteq X : |Y| = 26\} \Rightarrow |\Omega| = \binom{52}{26}$

Q3 52 cards into half (26 each)

$P(\underbrace{\text{Each half contains the same \# of red and black cards}}_{\parallel})$

$= P(\text{Each half contains 13 red cards and 13 black cards})$

$$= \frac{|E|}{|\Omega|} = \frac{\binom{26}{13}^2}{\binom{52}{26}}$$

$X = R \cup B$

$R := \{r_1, r_2, \dots, r_{26}\}$

$B := \{b_1, b_2, \dots, b_{26}\}$

$E = \{Y \subseteq X : |Y \cap R| = 13, |Y \cap B| = 13\} \subseteq \Omega$

$|E| = ? = \underbrace{\binom{26}{13}}_{\text{for } Y\cap R} \cdot \binom{26}{13} \to Y \cap B$

$$\frac{\binom{26}{13}^2}{\binom{52}{26}} = \frac{\left(\frac{26!}{13!\,13!}\right)^2}{\frac{52!}{26!\,26!}} = \frac{26!\,26!}{13!\,13!\,13!\,13!} \cdot \frac{26!\,26!}{52!}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$= \left(\frac{26!}{13!}\right)^4 \cdot \frac{1}{52!}$$

$$\underline{\underline{13}} = 13$$

$$26 = \underline{2 \cdot 13}$$

$$52 = \underline{2^2 \cdot 13}$$

$$\sim \left(\frac{\cancel{\sqrt{2\pi(26)}}\left(\frac{26}{e}\right)^{26}}{\cancel{\sqrt{2\pi(13)}}\left(\frac{13}{e}\right)^{13}}\right)^4 \cdot \frac{1}{\sqrt{2\pi(52)}\left(\frac{52}{e}\right)^{52}}$$

(Stirling)

$$= \left(\sqrt{2}\left(\frac{1}{e}\right)^{13} \underbrace{\frac{26^{26}}{13^{13}}}_{}\right)^4 \cdot \frac{1}{\sqrt{2\pi \cdot 52}\left(\frac{52}{e}\right)^{52}}$$

$$26^{13}\left(\frac{26}{13}\right)^{13} = 26^{13} \cdot 2^{13} = \underline{\underline{52^{13}}}$$

$$= \left(\sqrt{2} \cdot \frac{1}{e^{13}} \cdot 52^{13}\right)^4 \cdot \frac{e^{52}}{\sqrt{2\pi \cdot 52}\,(52)^{52}}$$

$$\boxed{\left(2^{\frac{1}{2}}\right)^4 = 2^2}$$

$$= \left(2^2 \cdot \frac{1}{\cancel{e^{52}}} \cdot \cancel{52^{52}}\right) \cdot \frac{\cancel{e^{52}}}{\sqrt{2\pi \cdot 52}\,\cancel{(52)^{52}}}$$

$$= \frac{4}{\sqrt{2\pi \cdot \underline{\underline{4}} \cdot 13}}$$

$$= \frac{2}{\sqrt{2\pi \cdot 13}}$$

$$= \frac{2}{\sqrt{26\pi}} \quad \checkmark$$

<u>Q4</u> 6n dice

Aim: $P(\text{each number appears exactly } n \text{ times}) = \frac{(6n)!}{(n!)^6}\left(\frac{1}{6}\right)^{6n}$

Total $= 6^{6n}$

$\Omega = \{(\overbrace{x_1, x_2, \dots, x_{6n}}^{\text{ordered tuple}}) : x \in \{1, 2, 3, \dots, 6\}\}$

$E = \{$6n-tuples ~~with~~ where i appears in n components for each $i = 1, 2, \dots, 6\}$

Exp: $(\underbrace{1, 1, \dots, 1}_{n}, \underbrace{2, \dots, 2}_{n}, \underbrace{3, \dots, 3}_{n}, \underbrace{4, \dots, 4}_{n}, \underbrace{5, \dots, 5}_{n}, \underbrace{6, \dots, 6}_{n}) \in E$

$$\therefore |E| = \frac{(6n!)}{(n!)^6}$$

↳ Each number repeated n times

$$\therefore \text{Prob.} = \frac{(6n)!}{(n!)^6}\cdot\left(\frac{1}{6}\right)^{6n} \sim \frac{\sqrt{2\pi\cdot 6n}\left(\frac{6n}{e}\right)^{6n}}{\left[\sqrt{2\pi\cdot n}\left(\frac{n}{e}\right)^{n}\right]^6}\cdot\frac{1}{6^{6n}}$$

$$= \frac{\sqrt{2\pi\cdot 6n}\left(\frac{\cancel{6n}}{\cancel{e}}\right)^{\cancel{6n}}}{(2\pi n)^3\left(\frac{\cancel{n^{6n}}}{\cancel{e^{6n}}}\right)}\cdot\frac{1}{\cancel{6^{6n}}}$$

$$= \frac{\sqrt{12\pi n}}{8\pi^3 n^3}$$

$$= \frac{\sqrt{12\pi}}{8\pi^3 n^{5/2}}$$

$$\therefore C = \frac{2\sqrt{3\pi}}{8\pi^3} = \frac{\sqrt{3\pi}}{4\pi^3} = \frac{\sqrt{3}}{4}\cdot\frac{1}{\pi^2\sqrt{\pi}}$$